

Diego Arnoldo Azuela Rosas

Engineer — Test & Systems

San Francisco, CA · (209) 401-7675 · diego.azuela@gmail.com · linkedin.com/in/diegoarnoldoazuelarosas · github.com/DiegoAzuela

PROFESSIONAL SUMMARY

Software engineer with 3+ years at Tesla building production Go and Python systems across high-voltage energy storage, humanoid robotics, and vehicle platforms. Primary author and maintainer of the shared Go instrument-driver and hardware-abstraction libraries that test systems across several factories depend on. Depth in UDS / ISO 14229 diagnostics and firmware download, CAN, gRPC services, concurrent state-machine architectures with strict timeout and fault semantics, and the SQL and analytics layer that turns test data into decisions.

PROFESSIONAL EXPERIENCE

Launch Engineer · Tesla, Inc.

Mar 2023 — Present · California

Energy Division — Software Development

- **Instrument Driver & HAL Platform (Golang):** primary author and top contributor of two shared Go libraries — a VISA/SCPI instrument-driver layer and a hardware-abstraction layer — consumed by test systems across multiple factories. Drivers for nine instrument vendors (Chroma, Keysight, Keithley, TDK, Itech, Magna, Siglent, NI, WAGO). Designed the 3-layer architecture and the SCPI command library for the Chroma hipot family; lifted instrument config into the shared HAL so new stations integrate by config, not code.
- **Megapack Thermal System (Golang, SCPI, ECU, Linux):** top software contributor on an HV test system covering Hipot, LV, HV and pneumatic actuation for coolant flow and leak detection; owned thermal and coolant-flow health monitoring. Co-authored the Pilz safety program and obtained safety buyoffs at 220V and 250V.
- **Safety Controller FVT (Golang, UDS, gRPC, CAN):** co-maintainer of a functional verification tester built as a state-machine application fronting a gRPC/protobuf bridge to CAN and UDS services. Implemented the full ISO 14229 flashing sequence — security access, bootloader entry by ECU reset with tester-present, application download under a dedicated context, genealogy writes by DID. Median-of-N sampling with CAN-MUX-aligned timing to prevent aliasing.
- **BMS Communication Station (Python, CAN):** CAN-bus polling station catching broken BMS chain-communication faults invisible to existing end-of-line testers.
- **Manufacturing Data & OEE Analytics (SQL, Power BI, Python):** architected a PLC-to-SQL diagnostic pipeline across the HV converter testers; maintain a large SQL analytics library covering failure-mode analysis, SCADA availability and OEE.
- **Telemetry Daemon (Linux, OPC UA):** daemon monitoring 200+ Siemens OPC UA nodes with automated fault alerting and recovery orchestration.
- **Humanoid Robotics — Optimus:** brought up the Charger end-of-line tester during NPI — ECU firmware debugging via CANape, DBC decoding and HEX analysis. Integrated AWS S3 as a centralized trace store for CAN logs.
- **Test platform tooling:** consolidated a duplicated test runner into one shared repository used across sites; built a Go parser generating safety documentation from Pilz PNOZ files, DBC revision diffing, and automated alerting.
- **Vehicles Division:** led BMS integration and wireless-charging test deployment at contract-manufacturer sites in Mexico and the USA.

RESEARCH EXPERIENCE

Research Intern · Harvard Medical School, Brigham and Women's Hospital

Jul 2021 — Feb 2022 · Cambridge, MA

- Developed Python control software for a 6-DOF robotic bioprinter using OpenCV depth sensing and computational-geometry path planning; validated deposition accuracy end-to-end. Published in *Advanced Materials Technologies*, 2024.

- Built an electrochemical-biosensor data-acquisition and signal-processing pipeline integrated into a microfluidic bioreactor for real-time stem-cell monitoring. Published in ACS Sensors, 2024.

EDUCATION

B.Sc. Mechatronics Engineering · Tecnológico de Monterrey (ITESM) <i>GPA 3.73</i>	2017 — 2023
B.Sc. Mechanical Engineering · Penn State University <i>GPA 3.75 · Dean's List</i>	2016 — 2017

TECHNICAL SKILLS

Languages	Golang, Python, SQL, C++, Shell, Bash, LabVIEW, TestStand, MATLAB
Software engineering	Go concurrency (goroutines, context, channels), gRPC & protobuf, REST integration, state-machine architecture, driver & library design, code review, semantic versioning & release tagging
Protocols	CAN bus, UDS / ISO 14229 (DIDs, security access, bootloader, FWDL), CANape, DBC, PCAN, HEX, OPC UA, SCPI & VISA, Modbus TCP, I2C, SPI, PLC, ECU firmware
Infrastructure	Linux (systemd), AWS S3, MySQL, Docker, CI/CD, pytest, Git, Ansible, Power BI, OpenCV
Methods	SPC, Gage R&R, fault injection, FMEA, NPI, DVT, PVT, V&V, root-cause analysis
Spoken	English (TOEFL 640), Spanish (native), French & Italian (intermediate)